

SQL Numeric Functions

ROUND, TRUNCATE, CEIL / CEILING, and FLOOR with practical examples

Function	Purpose
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1. ROUND(number, decimals)

Rounds a number to the specified number of decimal places using normal rounding rules.

SQL:

```
SELECT ROUND(12.345, 2);
```

Output: 12.35

2. ROUND(number)

Rounds a number to the nearest whole number when the decimal-place argument is omitted.

SQL:

```
SELECT ROUND(12.6);
```

Output: 13

3. TRUNCATE(number, decimals)

Cuts off digits after the specified decimal place without rounding the remaining value.

SQL:

```
SELECT TRUNCATE(12.349, 2);
```

Output: 12.34

4. CEIL(number) / CEILING(number)

Rounds a number upward to the nearest integer. CEILING() is an alternative name for CEIL() in MySQL.

SQL:

```
SELECT CEIL(12.1);
```

Output: 13

5. FLOOR(number)

Rounds a number downward to the nearest integer.

SQL:

```
SELECT FLOOR(12.9);
```

Output: 12

ROUND vs TRUNCATE

The main difference is that ROUND() changes the value when the next digit requires rounding, whereas TRUNCATE() simply removes digits after the requested decimal position.

SQL statement	Result
ROUND(12.349, 2)	12.35
TRUNCATE(12.349, 2)	12.34

Practical Example Using a Table

The following example creates a small product table, inserts sample values, and then uses all of the numeric functions on the price column.

1. Create the table

```
CREATE TABLE products (  
    product_id INT,  
    product_name VARCHAR(50),  
    price DECIMAL(10,3)  
);
```

2. Insert sample values

```
INSERT INTO products (product_id, product_name, price)  
VALUES  
(1, 'Keyboard', 1250.349),  
(2, 'Mouse', 799.756),  
(3, 'Monitor', 12499.925),  
(4, 'Headphones', 2499.499);
```

3. View the inserted data

```
SELECT * FROM products;
```

product_id	product_name	price
1	Keyboard	1250.349
2	Mouse	799.756
3	Monitor	12499.925
4	Headphones	2499.499

4. Apply the numeric functions

```
SELECT  
    product_name,  
    price,  
    ROUND(price, 2) AS rounded_price,  
    TRUNCATE(price, 2) AS truncated_price,  
    CEIL(price) AS ceiling_price,  
    FLOOR(price) AS floor_price  
FROM products;
```

Expected result:

product	price	ROUND	TRUNCATE	CEIL	FLOOR
Keyboard	1250.349	1250.35	1250.34	1251	1250
Mouse	799.756	799.76	799.75	800	799
Monitor	12499.925	12499.93	12499.92	12500	12499

Headphones	2499.499	2499.50	2499.49	2500	2499
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Quick Memory Guide

Function	Easy way to remember
ROUND()	Normal rounding.
TRUNCATE()	Cuts decimal digits without rounding.
CEIL()	Moves upward to the next integer.
CEILING()	Same idea as CEIL().
FLOOR()	Moves downward to the previous integer.